Corporate Finance

Lecture 2: Portfolio Choice and Diversification

Camilo Saldarriaga

Finance Department, ESCP Business School, 2026/2027

Plan

- Recap what we learned in Lecture 1: returns and risk
- Understand how combining assets reduces risk (diversification)
- Learn how to compute the risk and return of a portfolio

Key concepts

- Covariance and correlation between two stocks
- Diversification: which risk disappears in a portfolio and which does not

Recap of Lecture 1

Let's do a recap:

- The summation sign
- What a return is, and where it comes from
- Why returns compound, and what that does over a long horizon
- The geometric mean (CAGR) and the arithmetic mean
- Stock splits
- Variance and volatility
- The link between risk and return

Recap: the summation sign

- $\sum$ is a shortcut for a long addition. Below it: where to **start**; above it: where to **stop**; to its right: **what to add**.

$$\sum_{t=1}^{4} R_t = R_1 + R_2 + R_3 + R_4$$

- When we do not know how many terms there are, we give that number a name, say T, and stop there:

$$\sum_{t=1}^{T} R_t = R_1 + R_2 + \cdots + R_T$$

Recap: prices differ, returns allow us to compare

- A share of LVMH costs a few hundred euros, a share of Orange about ten. The prices alone say nothing about which was the better investment.
- The **return** puts every investment on the same scale: the percentage increase in value for every euro invested.

$$
\begin{aligned}
R_{t+1} &= \frac{\text{Wealth at the end}}{\text{Amount invested}} - 1 \\
&= \frac{\text{Wealth at the end} - \text{Amount invested}}{\text{Amount invested}} \\
&= \frac{\text{Change in value}}{\text{Amount invested}}
\end{aligned}
$$

- Example: you buy at EUR 50 and sell at EUR 55 one year later. Each euro became 1.10 euros, so the return is $55/50 - 1 = (55 - 50)/50 = 10\%$, whatever the size of your investment.

Recap: capital gains and dividends

- A stock pays you in two ways over the year: its price moves, and it may pay a dividend. The return counts both.

$$R_{t+1} = \frac{P_{t+1} + Div_{t+1} - P_t}{P_t} = \underbrace{\frac{P_{t+1} - P_t}{P_t}}_{\text{capital gain}} + \underbrace{\frac{Div_{t+1}}{P_t}}_{\text{dividend yield}}$$

- P_t is the price when you buy, P_{t+1} the price one year later, and Div_{t+1} the dividend received in between.
- Example: bought at 50, sold at 53, dividend of 2. Capital gain $3/50 = 6\%$, dividend yield $2/50 = 4\%$, total return 10%.

Recap: returns compound

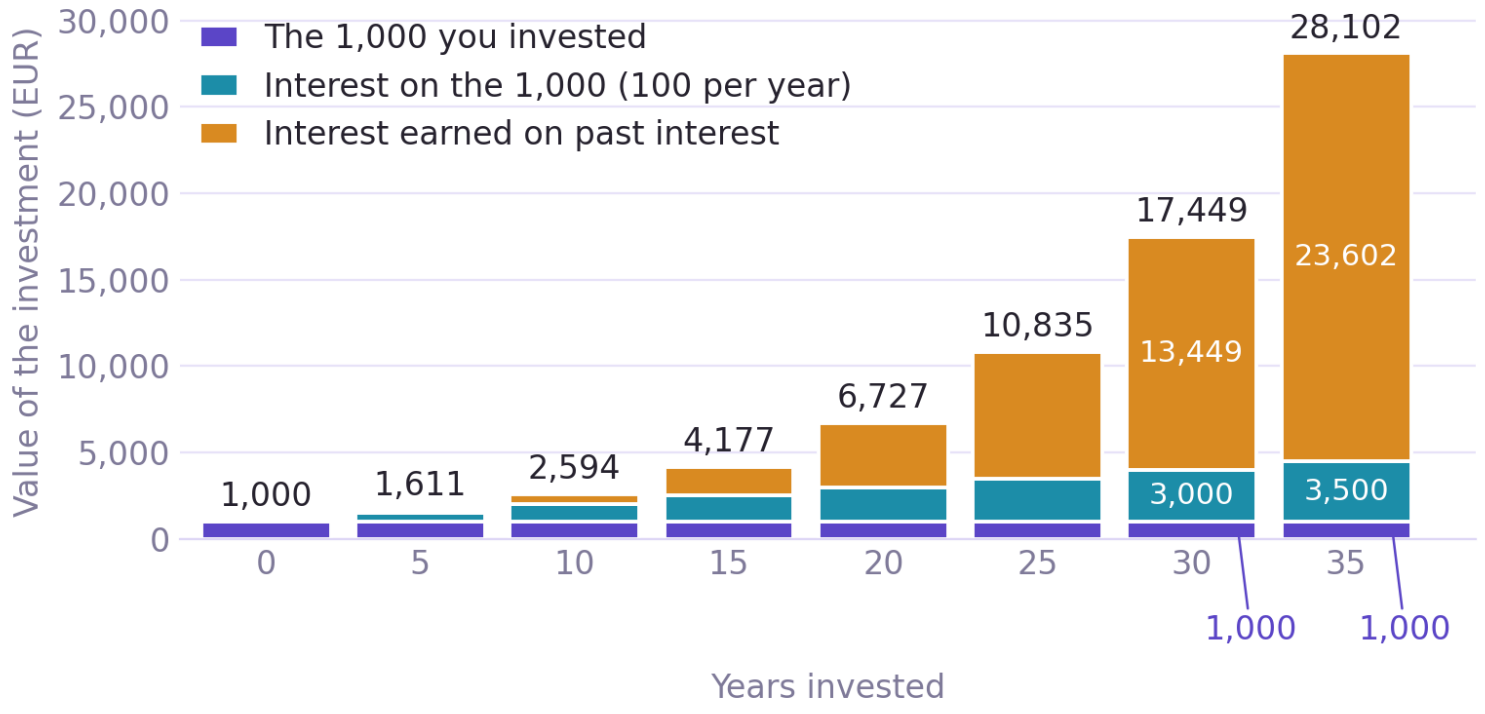
- After several periods, you earn interests not only on the principal, but also on the interests rates: you earn interests on the interests already earned. That is compounding.
- One euro becomes $(1 + R_1)$ after the first period, that amount then grows by $(1 + R_2)$, and so on. Over N periods:

$$1 + R_{1 \to N} = (1 + R_1)(1 + R_2) ... (1 + R_N)$$

- With $N = 4$ and returns of 6.1%, -5.32%, 6.72% and 8.93%:

$$R_{1 \to 4} = (1.061)(0.9468)(1.0672)(1.0893) - 1 = 16.7\%$$

Compounding: EUR 1,000 invested at 10%



Of the EUR 28,102 you end up with, only EUR 1,000 is your own money and EUR 3,500 is interest on it. The other EUR 23,602 is interest earned on past interest.

Recap: the geometric mean, or CAGR

- Since returns compound, the meaningful "average" return is the constant yearly return that would have produced the same final wealth. That is the geometric mean, also called the CAGR (compound annual growth rate).

$$\text{Geometric mean} = \left(\frac{\text{Wealth after } n \text{ years}}{\text{Amount invested}}\right)^{\frac{1}{n}} - 1$$

$$= \left((1+R_1)\,(1+R_2)\,...\,(1+R_n)\right)^{\frac{1}{n}} - 1$$

- Our four periods: $(1.167)^{\frac{1}{4}} - 1 = 3.94\%$ per period. Earning 3.94% four times in a row gives back exactly the same 16.7%.

Recap: stock splits

- In a stock split the company replaces each old share by several new ones. The price per share falls, but nothing of value has changed: you hold more, smaller slices of the same pie.
- A price drop after a split is therefore not a loss. Always compare the value of your whole position, not the price of a single share.
- Example: you hold 100 shares at EUR 150. The company does a 3:1 split: each share becomes 3, and the price drops to EUR 50. Looking only at the price, you would think you lost $50/150 - 1 = -66.7\%$. Looking at your position:

$$R = \frac{300 \times 50 - 100 \times 150}{100 \times 150} = \frac{15{,}000 - 15{,}000}{15{,}000} = 0\%$$

Recap: arithmetic mean and expected return

- The geometric mean describes what happened. To say what we **expect** next year, we use the arithmetic mean: add the returns, divide by how many there are.

$$\overline{R} = \frac{1}{T}\sum_{t=1}^{T} R_t = \frac{R_1 + R_2 + \cdots + R_T}{T}$$

- Our four returns give $(6.1 - 5.32 + 6.72 + 8.93)/4 = 4.11\%$, against a geometric mean of 3.94%.
- The arithmetic mean is always the larger of the two, unless every return is identical. It is our best guess of next period's return, and it is the one we will use from now on as the **expected return**, $E(R)$.

Recap: variance and volatility

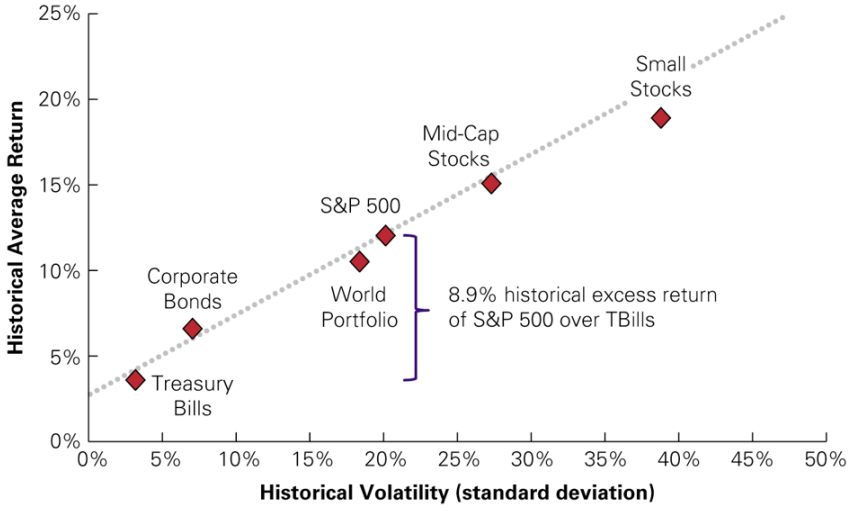
- Risk is about how far the actual return lands from its average. The variance measures it: take each year's distance from the mean, square it so that gains and losses both count, and average.

$$\text{Var}\,(R) = \frac{1}{T-1}\sum_{t=1}^{T}\left(R_t - \overline{R}\right)^2 \qquad \sigma_R = \sqrt{\text{Var}\,(R)}$$

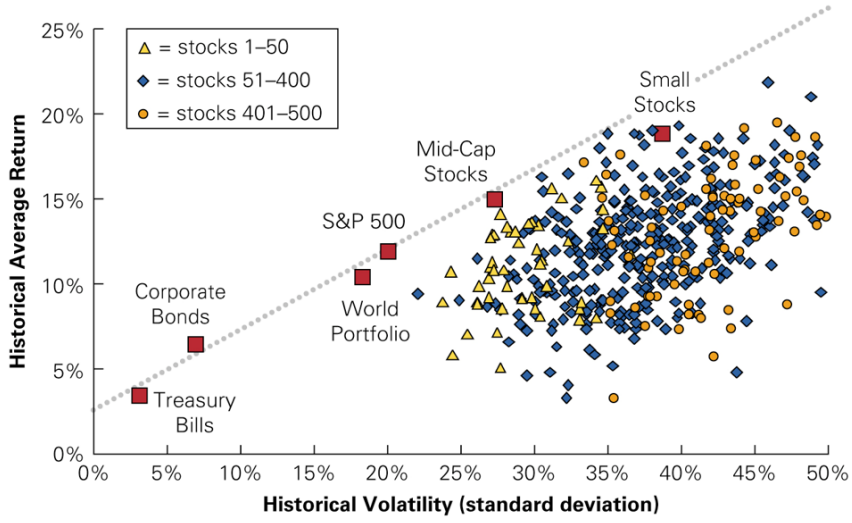
- The square root σ_R is the **volatility**. It is expressed in percentage points, so it can be read next to the mean return.
- Volatility is symmetric: it counts a good surprise exactly like a bad one.

Recap: risk and return

Large portfolios: clear link



Individual stocks: no clear link



- Big diversified portfolios line up neatly: more volatility, more return. Single stocks do not, and many of them carry a lot of risk for no extra return.
- Why? Part of the risk of a single stock disappears when it is held together with others, and is therefore not rewarded. **That is what this lecture is about.**

Beginning of lecture 2 content

Today we study the basis of portfolio choice. It can look abstract and distant at first, but it is the basis on which investment decisions are made every day, and on which a sector of trillions of euros operates: the financial sector.

Introduction

- Optimal portfolio choice:
  - How can we optimise risk and return by combining multiple assets?
  - Pioneered by Markowitz in his article "Portfolio Selection" published in the "Journal of Finance" in 1952.



Harry Markowitz, Nobel Memorial Prize 1990

Outline

A. Risk and return of a two-stock portfolio: the effect of diversification

- The content of this lecture mostly draws on the BDM textbook Chap. 10.5-6 and 11.1.

Which framework for choice under risk?

- Let's play a game
- Do you prefer:
  A. Win EUR 100 000 for sure.
  B. Participate in a lottery which yields EUR 0 with probability 0.5, and EUR 200 000 with probability 0.5?
- Distribution of expected gains:
  - $E(W_A) = 100000$
  - $E(W_B) = 0.5 \times 0 + 0.5 \times 200000 = 100000$
- According to the expected value criterion: indifference.
- And still: (in general) preference for the sure outcome! (i.e. aversion to risk). The mean-variance framework tries to address this.

How investors choose: two equivalent rules

We assume financial agents (investors) care about two things only: return and risk, measured by the variance.

Same return, different risk: they pick the less risky asset.

More formally: *for a given level of return, they minimize risk.*

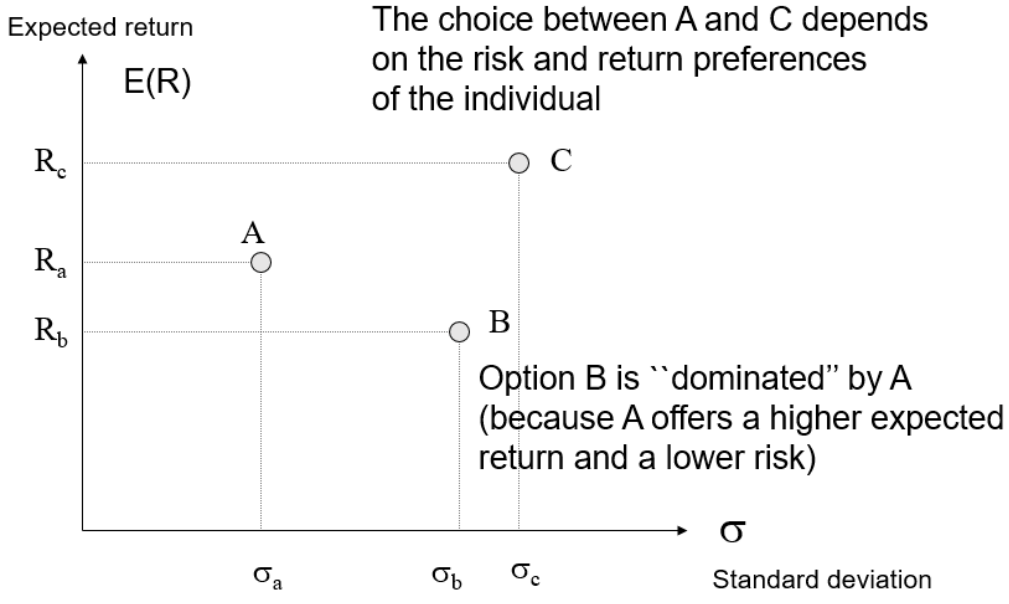
Same risk, different return: they pick the asset with the higher return.

More formally: *for a given level of risk, they maximize return.*

These two rules are **equivalent**: both describe the same investor, one who wants more return and less risk.

The mean-variance framework

- Every asset (or investment option) can be placed as a point on a chart: its **expected return** $E(R)$ on the vertical axis, and its **risk** on the horizontal axis.
- Risk is measured by the standard deviation σ, the square root of the variance, so both axes are in the same units (%).



Final wealth is a random variable

- When we invest, we do not know today how much we will end up with. The final wealth W is a **random variable**: it can take several values.
- What we can do is estimate its **probability distribution**: how likely or unlikely each end result is. In the lottery of the game: $W = 0$ with probability 0.5, $W = 200{,}000$ with probability 0.5.
- Two numbers summarize a distribution. They are called its **first two moments**:
  - the mean $E(W)$: what we get on average;
  - the variance $\text{Var}(W)$: how spread out the possible results are.
- The sure EUR 100,000 and the lottery have the same $E(W)$, but the lottery has a much larger $\text{Var}(W)$.

Mean-variance utility

- The mean-variance idea is formalized with a **utility function**: a score of how satisfied an investor is with a risky investment. We assume each investor picks the investment with the highest utility.
- The utility depends on the first two moments of wealth: $U(E(W), \text{Var}(W))$. It goes up with the expected wealth $E(W)$ and down with the variance of wealth $\text{Var}(W)$.
- A common specification is

$$U(E(W), \text{Var}(W)) = E(W) - \frac{\gamma}{2}\text{Var}(W),$$

  where $\gamma > 0$ indicates how much investors dislike variance in comparison with mean returns.
- More risk aversion or lower risk tolerance refer to higher values of γ.

A. Risk and return of a two-stock portfolio: the effect of diversification

- You can invest in two stocks:
- The appropriate combination of stocks can produce an investment with lower risk. **This is extremely powerful and important!!!**
- Intuition investing in stocks that co-move in opposite directions reduces the risk
- EUR 50 invested in Umbrella Inc and EUR 50 in Sun Umbrella Inc.

Two risky stocks: a more realistic example of diversification (with fictitious stocks)

	Stock Returns			Portfolio Returns	
Year	North Air	West Air	Tex Oil	$1/2R_N + 1/2R_W$	$1/2R_W + 1/2R_T$
2013	21%	9%	−2%	15.0%	3.5%
2014	30%	21%	−5%	25.5%	8.0%
2015	7%	7%	9%	7.0%	8.0%
2016	−5%	−2%	21%	−3.5%	9.5%
2017	−2%	−5%	30%	−3.5%	12.5%
2018	9%	30%	7%	19.5%	18.5%
Average Return	10.0%	10.0%	10.0%	10.0%	10.0%
Volatility	13.4%	13.4%	13.4%	12.1%	5.1%

But first, let's face "3 monsters" that we will need here

$$\sigma_{X,Y} \qquad \rho_{X,Y} \qquad \beta_{Y,X}$$

Covariance **Correlation** **Beta**

Three ways of saying the same thing: how two stocks move together

Monster 1: the covariance

- The covariance asks one question: **when one stock does better than usual, does the other one do better than usual too?**
- For each year, measure how far stock i landed from its own average return, do the same for stock j, and multiply the two distances. Then average over the T years:

$$\text{Cov}\left(R_i, R_j\right) = \frac{1}{T-1} \sum_{t=1}^{T} \left(R_{it} - \overline{R}_i\right)\left(R_{jt} - \overline{R}_j\right)$$

- R_{it} is the return of stock i in year t and $\overline{R}_i$ its average return over the whole period.

Monster 1: reading the covariance

- **Both above their average, or both below**: the two distances have the same sign, their product is positive. Years like this push the covariance up.
- **One above while the other is below**: the product is negative and pushes the covariance down.
- So a **positive** covariance means the two stocks tend to move together, a **negative** one that they tend to move in opposite directions, and a covariance near zero that there is no systematic link.
- Only the sign is easy to read. The size depends on the units, which is why we rescale it into the correlation.

Monster 1: the covariance with scenarios

- Sometimes we do not have past years, but a list of possible scenarios: a boom, a normal year, a recession, each with its probability.
- The idea does not change. We weight each scenario by its probability p_i instead of counting every year equally:

$$\sigma_{X,Y} = \sum_i p_i \left(x_i - E\left(X\right)\right)\left(y_i - E\left(Y\right)\right)$$

- x_i and y_i are the returns of the two stocks in scenario i, and $E(X)$, $E(Y)$ their expected returns.

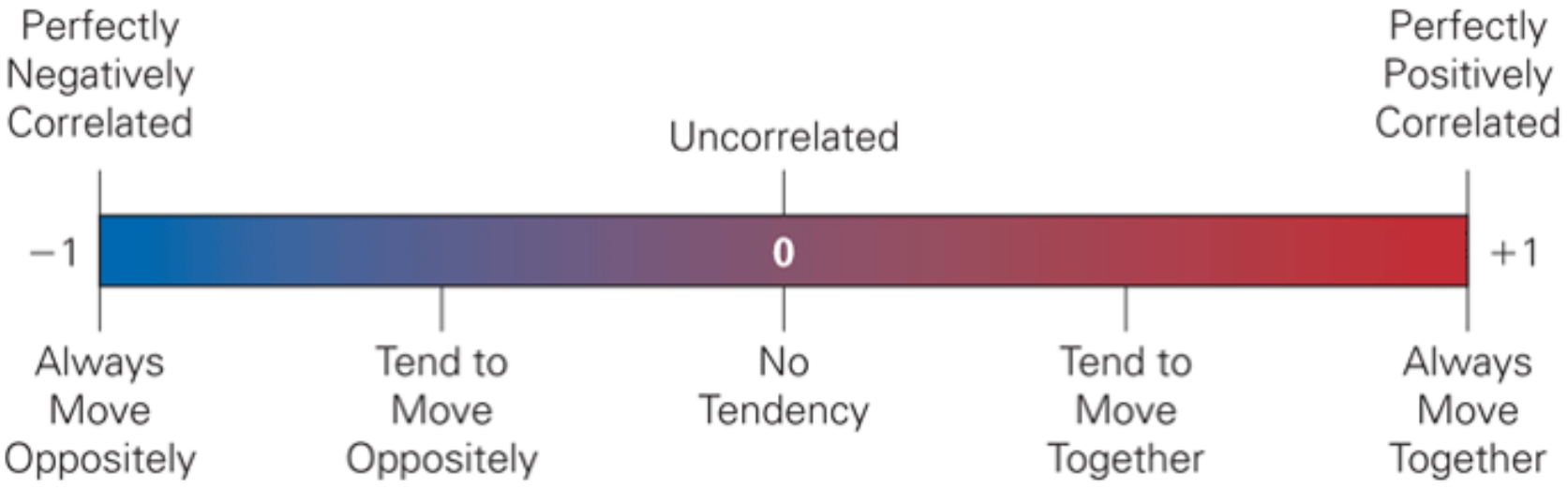
Monster 2: the correlation

- The covariance tells us the direction, but its size is not comparable across pairs of stocks. The correlation fixes that: divide the covariance by the two volatilities.

$$\rho_{X,Y} = \text{Corr}\left(R_i, R_j\right) = \frac{\text{Cov}\left(R_i, R_j\right)}{\sigma_{R_i}\sigma_{R_j}}$$

- The units cancel, so the result is always a number between -1 and $+1$:
  - $+1$: the two always move together, in perfect step.
  - 0: no systematic link between them.
  - -1: when one goes up, the other goes down, every time.
- It is the same information as the covariance, on a scale we can read.

Interpretation of the correlation



The correlation measures how returns move in relation to each other.

Monster 3: the beta

- The beta answers a different question: **when X moves by 1%, by how much does Y move on average?** It is the covariance divided by the variance of X, the one that does the moving:

$$\beta_{Y,X} = \frac{\text{Cov}\,(X,Y)}{\text{Var}\,(X)} = \frac{\sigma_{X,Y}}{\sigma_X^2}$$

- If X is the whole market, the beta says how sensitive the stock Y is to the market: a beta of 1.5 means that a market rise of 1% comes with a rise of about 1.5% for that stock, and a fall of 1% with a fall of about 1.5%.
- A beta above 1 amplifies the market, below 1 dampens it, and a negative beta moves against it.
- Beta comes back as the central measure of risk in Lecture 4 (the CAPM).

The three monsters side by side

Measure	Formula	What it tells us	Range
Covariance	$\sigma_{X,Y}$	Do they move together?	any number
Correlation	$\sigma_{X,Y}/(\sigma_X\sigma_Y)$	How strongly, on a fixed scale	-1 to $+1$
Beta	$\sigma_{X,Y}/\sigma_X^2$	How much Y moves when X moves 1%	any number

- All three are built from the same covariance. They only differ in what we divide it by.
- Covariance and correlation always have the same sign: if two stocks move together, both are positive.

Computing the covariance and correlation between two stocks

Year	Deviation from Mean $(R_N - \overline{R}_N)$	$(R_W - \overline{R}_W)$	$(R_T - \overline{R}_T)$	North Air and West Air $(R_N - \overline{R}_N)(R_W - \overline{R}_W)$	West Air and Tex Oil $(R_W - \overline{R}_W)(R_T - \overline{R}_T)$
2013	11%	−1%	−12%	−0.0011	0.0012
2014	20%	11%	−15%	0.0220	−0.0165
2015	−3%	−3%	−1%	0.0009	0.0003
2016	−15%	−12%	11%	0.0180	−0.0132
2017	−12%	−15%	20%	0.0180	−0.0300
2018	−1%	20%	−3%	−0.0020	−0.0060
			Sum $= \sum_t (R_{i,t} - \overline{R}_i)(R_{j,t} - \overline{R}_j) =$	0.0558	−0.0642
Covariance:			$Cov(R_i, R_j) = \frac{1}{T-1}\text{Sum} =$	0.0112	−0.0128
Correlation:			$Corr(R_i, R_j) = \frac{Cov(R_i, R_j)}{SD(R_i)SD(R_j)} =$	0.624	−0.713

Historical annual volatilities and correlations for selected stocks

	Microsoft	HP	Alaska Air	Southwest Airlines	Ford Motor	Kellogg	General Mills
Volatility (Standard Deviation) Correlation with	32%	36%	36%	31%	47%	19%	17%
Microsoft	1.00	0.40	0.18	0.22	0.27	0.04	0.10
HP	0.40	1.00	0.27	0.34	0.27	0.10	0.06
Alaska Air	0.18	0.27	1.00	0.40	0.15	0.15	0.20
Southwest Airlines	0.22	0.34	0.40	1.00	0.30	0.15	0.21
Ford Motor	0.27	0.27	0.15	0.30	1.00	0.17	0.08
Kellogg	0.04	0.10	0.15	0.15	0.17	1.00	0.55
General Mills	0.10	0.06	0.20	0.21	0.08	0.55	1.00

Properties of the covariance and variance (reminder)

- The covariance is bilinear:

$$\mathsf{Cov}\left[R_1 + R_2, R_3\right] = \mathsf{Cov}\left[R_1, R_3\right] + \mathsf{Cov}\left[R_2, R_3\right]$$
$$\mathsf{Cov}\left[mR_2, R_3\right] = m\mathsf{Cov}\left[R_2, R_3\right]$$

- The variance is a special case of the covariance

$$\mathsf{Var}\left[R_1\right] = \mathsf{Cov}\left[R_1, R_1\right]$$
$$\mathsf{Var}\left[mR_1\right] = m\mathsf{Cov}\left[R_1, mR_1\right] = m^2\mathsf{Cov}\left[R_1, R_1\right] = m^2\mathsf{Var}\left[R_1\right]$$

Computing the variance and standard deviation of a portfolio: the case with two stocks

- The return of the portfolio is $R_p = x_1 R_1 + x_2 R_2$.
- x_1 and x_2 are the **weights**: the fraction of your money in each stock, adding up to 1. EUR 30 of every EUR 100 in stock 1 and EUR 70 in stock 2 means $x_1 = 0.30$ and $x_2 = 0.70$, so each return counts in proportion to the money behind it.
- We get

$$\begin{aligned}
\mathsf{Var}\left(R_p\right) &= \mathsf{Cov}\left(R_p, R_p\right) = \mathsf{Cov}\left(x_1 R_1 + x_2 R_2, x_1 R_1 + x_2 R_2\right) \\
&= x_1^2 \mathsf{Var}\left(R_1\right) + x_2^2 \mathsf{Var}\left(R_2\right) + 2 x_1 x_2 \mathsf{Cov}\left(R_1, R_2\right) \\
&= x_1^2 \mathsf{Var}\left(R_1\right) + x_2^2 \mathsf{Var}\left(R_2\right) + 2 x_1 x_2 \sigma_{R_1} \sigma_{R_2} \mathsf{Corr}\left(R_1, R_2\right)
\end{aligned}$$

- The volatility of the portfolio is

$$\sigma_{R_p} = \sqrt{\mathsf{Var}\left(R_p\right)}.$$

The risk of a portfolio with two stocks

- The risk of a portfolio with two stocks is **not** the sum of the individual stock risks.
- It is the sum of the individual stock risks, plus the comovement of the stocks:

$$\text{Var}\left(R_p\right) = \underbrace{x_1^2\text{Var}\left(R_1\right) + x_2^2\text{Var}\left(R_2\right)}_{\text{individual risks}} + \underbrace{2x_1x_2\text{Cov}\left(R_1, R_2\right)}_{\text{comovement}}$$

- If they co-move in different directions, the aggregated risk is smaller.
- If they co-move in the same direction, the aggregated risk is greater.

Diversification and correlation

σ_p is the volatility of the portfolio, and ρ_{12} the correlation between the two stocks.

$\rho_{12} = 1 \Rightarrow \sigma_p = x_1\sigma_1 + x_2\sigma_2$ The risk of the portfolio is the average risk.

$\rho_{12} < 1 \Rightarrow \sigma_p < x_1\sigma_1 + x_2\sigma_2$ The risk of the portfolio is smaller than the average risk due to the effect of diversification.

$\rho_{12} = -1 \Rightarrow \sigma_p = |x_1\sigma_1 - x_2\sigma_2|$ The negative comovement of the assets reduces the risk of the portfolio. The risk of the portfolio is the absolute value of the difference between the weighted risks.

The volatility of a portfolio of N stocks

- The portfolio return is

$$R_p = x_1 R_1 + \cdots + x_N R_N = \sum_{i=1}^{N} x_i R_i.$$

- The variance is

$$\begin{aligned}
\text{Var}\left[R_p\right] &= \text{Cov}\left[R_p, R_p\right] = \sum_{i=1}^{N} x_i \text{Cov}\left[R_i, R_p\right], \\
&= \sum_{i=1}^{N} x_i \text{Cov}\left[R_i, \sum_{j=1}^{N} x_j R_j\right] = \sum_{i=1}^{N} x_i \sum_{j=1}^{N} x_j \text{Cov}\left[R_i, R_j\right] \\
&= \sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j \text{Cov}\left[R_i, R_j\right].
\end{aligned}$$

- **Don't panic**: We will not use this formula for more than two stocks.

Can a weight be negative?

- So far every weight was between 0 and 1: you put part of your money in each stock.
- A weight can also be **negative**. It means you have sold a stock you do not own. This is a **short sale**, or "shorting" the stock.
- How is that possible? You borrow the share from someone (in practice your broker), sell it today, and buy it back later to give it back. You hold the cash now and owe one share later.
- You gain if the price falls: sell at 100, buy back at 60, return the share, and keep 40. You lose if the price rises: buying back at 150 costs you 50.

Shorting: an everyday picture

- A friend lends you a soccer match ticket and says "give it back next month".
- You think tickets will be cheaper next month, so you sell it today for EUR 100.
- Next month tickets cost EUR 60. You buy one, hand it back to your friend, and keep the EUR 40 difference. You made money on a ticket you never owned.
- If tickets had risen to EUR 150 instead, you would still have to buy one to return it, and lose EUR 50.
- Two lessons: a short position **wins when the price goes down and loses when it goes up**, and there is no ceiling on the loss, because a price can keep rising.

Negative weights in the portfolio formula

- You have EUR 100. You short EUR 50 of stock 1 and put all the cash you now hold, your EUR 100 plus the EUR 50 raised, into stock 2.
- The weights are $x_1 = -50/100 = -0.5$ and $x_2 = 150/100 = 1.5$. They still add up to 1, since the money always comes from somewhere.
- The formulas do not change:

$$R_p = x_1 R_1 + x_2 R_2 = -0.5\, R_1 + 1.5\, R_2$$

- If stock 1 falls, the negative weight turns that fall into a gain for the portfolio. Lecture 3 uses negative weights to widen the set of portfolios we can build.

Worked example: two stocks, four years

We observe the yearly returns of two stocks over four years:

Year	Stock A	Stock B
1	12%	1%
2	-4%	9%
3	10%	-3%
4	2%	5%

- Notice that they tend to move in opposite directions: A's best years are B's worst.
- We compute, step by step: the mean returns, the variances and volatilities, the covariance, the correlation, the beta, and then the return and risk of a portfolio that puts half of the money in each stock.
- To keep the arithmetic simple, we work in percentage points throughout.

Step 1: mean returns

- Add the four returns and divide by four:

$$\overline{R}_A = \frac{12 - 4 + 10 + 2}{4} = \frac{20}{4} = 5\%$$

$$\overline{R}_B = \frac{1 + 9 - 3 + 5}{4} = \frac{12}{4} = 3\%$$

- These are our estimates of the expected return of each stock: on a typical year, A earns 5% and B earns 3%.

Step 2: variances and volatilities

- For each year, take the distance to the mean, square it, then add up:

Year	$R_A - 5$	squared	$R_B - 3$	squared
1	7	49	-2	4
2	-9	81	6	36
3	5	25	-6	36
4	-3	9	2	4
Sum		**164**		**80**

- Divide by $T - 1 = 3$ and take the square root:

$$\text{Var}(R_A) = \frac{164}{3} = 54.7 \qquad \sigma_A = \sqrt{54.7} = 7.4\%$$

$$\text{Var}(R_B) = \frac{80}{3} = 26.7 \qquad \sigma_B = \sqrt{26.7} = 5.2\%$$

Step 3: covariance

- Same distances as before, but now multiply A's distance by B's, year by year:

Year	$R_A - 5$	$R_B - 3$	product
1	7	-2	-14
2	-9	6	-54
3	5	-6	-30
4	-3	2	-6
Sum			**-104**

$$\text{Cov}(R_A, R_B) = \frac{-104}{3} = -34.7$$

- Every product is negative: in each year, when one stock is above its average the other is below. The two stocks move in opposite directions.

Step 4: correlation and beta

- Correlation: divide the covariance by the two volatilities.

$$\rho_{A,B} = \frac{-34.7}{7.4 \times 5.2} = \frac{-34.7}{38.2} = -0.91$$

Close to -1: the two stocks move almost perfectly against each other.

- Beta of B with respect to A: divide the covariance by the variance of A, the stock that does the moving.

$$\beta_{B,A} = \frac{-34.7}{54.7} = -0.63$$

When A goes up by 1%, B goes down by about 0.63% on average.

Step 5: return of the 50/50 portfolio

- Put half of the money in each stock: $x_A = x_B = 0.5$.
- Year by year, the portfolio earns the average of the two returns:

Year	Stock A	Stock B	Portfolio
1	12%	1%	6.5%
2	-4%	9%	2.5%
3	10%	-3%	3.5%
4	2%	5%	3.5%

- Its mean return is $(6.5 + 2.5 + 3.5 + 3.5)/4 = 4\%$, exactly what the formula gives:

$$\overline{R}_p = x_A \overline{R}_A + x_B \overline{R}_B = 0.5 \times 5 + 0.5 \times 3 = 4\%$$

Step 6: risk of the 50/50 portfolio

- With the formula from this lecture:

$$\text{Var}(R_p) = 0.5^2 \times 54.7 + 0.5^2 \times 26.7 + 2 \times 0.5 \times 0.5 \times (-34.7)$$

$$= 13.7 + 6.7 - 17.3 = 3.0 \qquad \sigma_p = \sqrt{3.0} = 1.7\%$$

- Check directly on the portfolio returns: the distances to the 4% mean are 2.5, -1.5, -0.5, -0.5; their squares add up to 9, and $9/3 = 3.0$. Same answer.
- **Takeaway**: alone, the stocks have volatilities of 7.4% and 5.2%. Together, 1.7%. The negative covariance term removed most of the risk while the return stayed a plain average of the two. That is diversification at work.